X Hour and YY minutes

THE UNIVERSITY OF MANCHESTER

EART1200 Toolkit – Maths and Physics

?? June 2024

Time**: ??:?? - ??:??**

Answer ALL questions in the booklets provided.

Clearly label which question each of your answers corresponds to. All writing must be LEGIBLE. Illegible documents cannot be marked.

There are 100 marks available on this paper. The number of marks for each question is shown and may be used as a guide to how much time you should spend on each.

You may wish to use labelled sketches to illustrate your answers where appropriate.

Please show all of your working to get to an answer. This is used to mark correct methodologies.

The use of calculators **is permitted**, as long as they cannot store text, perform algebra and have no graphing capability.

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The chain rule:
To obtain the derivative of a function f , which can be written as another function, g so we can use the common laws of differentiation, we chain together the following operations:

$$\frac{df}{dx} = \frac{df}{dg} * \frac{dg}{dx}$$

The product rule:
To obtain the derivative of a function, f, which appears to be the product of two functions u and v, we can deal with each separately using the following:

$$\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

The quotient rule:
To obtain the derivative of a function f which can be written as function u divided by function v, we can use the following:

$$\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Common differentials:

$$y = x^n; \frac{dy}{dx} = nx^{n-1}$$
$$y = e^{f(x)}; \frac{dy}{dx} = \frac{df}{dx}e^{f(x)}$$
$$y = \ln(x); \frac{dy}{dx} = \frac{1}{x}$$
$$y = \sin(x); \frac{dy}{dx} = \cos(x)$$
$$y = \cos(x); \frac{dy}{dx} = -\sin(x)$$

Common integrals:

$$\frac{dy}{dx} = x^n; y = \frac{x^{n+1}}{n+1} + constant$$
$$\frac{dy}{dx} = \frac{1}{x}; y = \ln(x) + constant$$
$$\frac{dy}{dx} = \cos(x); y = \sin(x) + constant$$
$$\frac{dy}{dx} = \sin(x); y = -\cos(x) + constant$$

1) Select TRUE or FALSE to the following statements about differentiation.
   (a) If $f(x) = \cos(2x)$ then $\frac{df}{dx} = 2\sin(2x)$
   (b) If $f(x) = e^{-2x}$ then $\frac{df}{dx} = \frac{-e^{-2x}}{2}$
   (c) If $f(x) = \ln(2x)$ then $\frac{df}{dx} = \frac{1}{2x}$
   (d) If $f(x) = 5x^n$ then $\frac{df}{dx} = 5nx^n$
   (e) If $f(x) = \sin(x)$ then $\frac{d^2f}{dx^2} = -\sin(x)$
   [10 marks]

2) Select the correct integral, $\int f(x)\,dx$, of the following function $f(x)$ from the list given below. Please show your working.
   [10 marks]

$$f(x) = 4x^3 - 2x + 5$$

$$a)\ x^4 - x^2 + 5x + C$$
$$b)\ x^4 - 2x^2 + 5x + C$$
$$c)\ x^4 - x^2 + 5 + C$$
$$d)\ 3x^4 - x^2 + 5$$

3) Using Newtons method, find the root of the following function

$$f(x) = e^x - x^2$$

   correct to 3 decimal places. You may use a starting guess of $x_0 = -0.6$. Please note that we can write Newtons formula, generally, as:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

   where $f'(x)$ is the derivative of $f(x)$.
   [10 marks]

4) Integrate the following expression and give your answer to 3 decimal places
   [10 marks]

$$\int_{-3}^{4} f^2 - e^{3f}\,df$$

5) A gas leak is emitting Butane at a variable rate. The concentration equation is given below, where G is given in micrograms per cubic metre and the x-axis given in minutes.

   The associated equation for G is also given as.

$$G = 3 + 14t - 5t^2$$

   Find the time at which the maximum concentration of emitted gas occurs. [10 marks]

6) Differentiate the following expressions with respect to *x*. Simplify your answer as much as possible.
   (a) $f = \cos(x)\, e^{-5x}$
   (b) $g = x/\sqrt{x^2 + 4}$
   (c) $y = e^{-x^2}/x$
   [10 marks]

7) A fuel tank has a slow releasing leak. We want to study the volume of fuel in the tank over time t which can be described using the following equation:

$$V(t) = \frac{V_0}{10}(t^2 + 10)\exp\left(-\frac{t}{2}\right)$$

   where $V_0 = 10$ is the volume of fuel (in m^3) and $0 < t < 24$ (in hours). What is the rate of the leak at 2 hours? Determine the rate analytically, but also compare to a rate calculated numerically using central differences method with a time step Δt of 30 minutes. Estimate the percentage error of the numerical result. [10 marks]

8) A body of mass m kilograms rests on a slope that has an angle of inclination of 30 degrees.

   (a) Calculate the speed of this body after 10 seconds, assuming the initial velocity is 0 $m.s^{-1}$ and the earths gravitation constant $g = 9.8m.s^{-2}$. For this problem you need to write all the forces acting on the body in the form of Newtons second law.

   $$F = ma = m\frac{dv}{dt}$$

   Where F is the total force acting on a body of mass m, leading to a rate of change in velocity $\frac{dv}{dt}$. You can then integrate this equation and evaluate it the given initial conditions [4 marks].

   (b) At time t=10, a force is suddenly applied to the body but opposite to the direction of travel. We are told this force is directly proportional to the mass m via a coefficient α, where $\alpha = 8.5$. The new force is thus given as $F = \alpha m$. Draw a schematic diagram that displays all the forces now acting on the body [2 marks] and determine how long it takes for the vehicle to stop, to 2 decimal places [4 marks].

9) The following equation described the rate of change in a population P

$$\frac{dP}{dt} = \alpha P - \beta P$$

Where α is a coefficient that determines a contribution according to births, whilst β a coefficient that determines the contribution from natural deaths.

   a) Integrate equation xx to arrive at the following equation that describes how P varies with time, if P_o represents the population value when $t = 0$:

   $P = P_o e^{t(\alpha-\beta)}$ [4 marks]

   b) We are now told the birth rate is constant, such that:

   $$\frac{dP}{dt} = \psi - \beta P$$

   Where ψ represents this constant. To integrate this expression, you will need to use integration using a substitution, creating a new variable equal to $(\psi - \beta P)$, to arrive at the following expression:
   [6 marks]

   $$\therefore P = \frac{1}{\beta}\left(\psi - (\psi - \beta P_0)e^{\beta t}\right)$$

10) The hydrostatic equation allows us to calculate the variation in pressure P with height z in the atmosphere:

$$\frac{dP}{dz} = -\frac{P}{R_a\left(T_{surface} - \alpha\, z\right)}g$$

Where $R_a = 287 Jkg^{-1}K^{-1}$.We have been told that the value for α could be either 0.01 or 0.02. We want to know what difference this makes in the final pressure at a height of 1400m.

a) Integrate the hydrostatic equation using a substitution, when $\alpha = 0.01$. This should lead you to derive the following value for the pressure at 1400m
[6 marks]

$$P = 100000 \exp\left(\frac{g}{0.01 \times 287} \ln \quad \frac{276}{290}\right) \cong 84454\ Pa$$

b) What is the percentage difference in pressure at 1400m when we change $\alpha = 0.01$ to $\alpha = 0.02$. Use 5 significant digits for the value of P at 1400m as we have done for part 'a)' and 1 decimal place for the percentage difference
[4 marks]